Constantin Alexander Auga

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Google Scholar | GitHub | Website | LinkedIn

Research Interests

My research spans computer vision, audio, and multimodal learning, including generative modeling, self-supervised learning, and multimodal model evaluation. I am particularly interested in how learned representations support perception, prediction, and world understanding, with a focus on multimodal video learning and dynamic environments.

Education

Hasso Plattner Institute, University of Potsdam Oct 2024 – Jun 2027 (expected)
M.Sc. IT Systems Engineering | Current grade: **1.1** (1.0 best)

University of Technology Sydney Feb – Jun 2027 (upcoming)
Selected for an exchange semester

University of Hamburg Oct 2021 – Sep 2024
B.Sc. Computer Science | Final grade: 2.0 (top 26% of cohort)

Publications and Preprints

Beyond Accuracy: Benchmarking Cross-Task Consistency in Unified Multimodal Models
WACV 2027 – Accepted [Paper] [Project] [Code]
Weixing Wang*, Liudvikas Zekas*, Anton Hackl*, **Constantin Alexander Auga***, Parisa Shahabinejad, Jona Otholt, Antonio Rueda-Toicen, Christoph Meinel, Gerard de Melo.
* Equal contribution.

- Co-developed XTC-Bench (31,530 scene-graph facts across 2,000 images) and cross-task agreement metrics; evaluated unified multimodal models, revealing consistent hallucinations missed by accuracy alone.

TargetSEC: Plug-and-Play In-the-Wild Speech Emotion Conversion via Arousal-Conditioned Latent Style Diffusion
ICASSP 2027 – Under review [Paper] [Project] [Code]
Constantin Alexander Auga (sole author)

- Developed an arousal-conditioned latent diffusion approach for duration-preserving emotion conversion; outperformed non-duration baselines on MSP-Podcast while maintaining speech naturalness.

Research Experience

M.Sc. Thesis Research – Hearing Beyond the Frame Ongoing
Hasso Plattner Institute, University of Potsdam | Supervisor: Gerard de Melo

- Building a JEPA-inspired framework that combines pretrained visual and audio representations to predict latent future states for action anticipation.
- Created an automatically filtered, human-verified benchmark with silent- and shuffled-audio controls; analyzed where auditory cues improve prediction.

Research Assistant – Max Delbrück Center Sep 2025 – Present
Continuous convolutional neural networks (CCNNs)

- Studied resolution robustness in 2D/3D models, designed evaluation metrics, and maintained reproducible experiments; contributed to aQQua, a tool for interpreting complex embedding spaces.

Research Assistant – German Aerospace Center (DLR) Sep 2023 – Sep 2024
Machine learning for hydrogen-powered aviation

- Supported ML-based leakage localization; built real-time sensor pipelines and experimental heatmap visualizations.

Awards and Academic Service

1st Place, Jaguar Re-Identification Challenge (Kaggle) Mar 2026
Ranked 1st of 370+ competitors with a DINOv3-based visual re-identification system; improved the baseline by 25%.

Student Representative, Hasso Plattner Institute Nov 2025 – Present
Served on official selection committees for two professorships.

Technical Skills and Languages

Programming: Python, C++, C#, Java, JavaScript, SQL **ML/tools:** PyTorch, Hugging Face, NumPy, pandas, scikit-learn, Git, Linux, ROS
Languages: German (native); English (TOEFL iBT 103, CEFR C1); Portuguese, Spanish, Georgian (basic)